HWK4: Welford's Recurrence Formulas and Sampling Mean/Variance

1 Exercise 1 – Welford's recurrence formulas

Let $x_1, \ldots, x_n$ be observations, and define the running mean and sum of squared deviations by

$$\bar{x}_n = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad SS_n = \sum_{i=1}^{n}(x_i - \bar{x}_n)^2.$$

For the new observation x_n, write

$$\delta_n = x_n - \bar{x}_{n-1}.$$

The usual recurrence for the mean is

$$\boxed{\bar{x}_n = \bar{x}_{n-1} + \frac{\delta_n}{n}}.$$

To obtain the recurrence for SS_n, first note that

$$x_i - \bar{x}_n = (x_i - \bar{x}_{n-1}) - (\bar{x}_n - \bar{x}_{n-1}) = (x_i - \bar{x}_{n-1}) - \frac{\delta_n}{n}.$$

Since $\sum_{i=1}^{n-1}(x_i - \bar{x}_{n-1}) = 0$, expanding the first $n-1$ squared deviations gives

$$\begin{aligned} SS_n &= \sum_{i=1}^{n-1}\left[(x_i - \bar{x}_{n-1}) - \frac{\delta_n}{n}\right]^2 + (x_n - \bar{x}_n)^2 \\ &= SS_{n-1} + \frac{n-1}{n^2}\delta_n^2 + \left(\frac{n-1}{n}\delta_n\right)^2 \\ &= SS_{n-1} + \frac{n-1}{n}\delta_n^2. \end{aligned}$$

Equivalently, because $x_n - \bar{x}_n = \frac{n-1}{n}\delta_n$,

$$\boxed{SS_n = SS_{n-1} + \delta_n(x_n - \bar{x}_n)}.$$

This is Welford's sum-of-squares update (often denoted M_2). The unbiased sample variance is therefore

$$\boxed{s_n^2 = \frac{SS_n}{n-1}}, \qquad n \geq 2.$$

For comparison, dividing by n instead would give the variance with divisor n, which is biased as an estimator of the population variance.

Minimal JavaScript implementation

The state consists only of the count, running mean, and $M_2 = SS$:

```
function add(x) {
  const delta = x - mean;
  mean += delta / ++count;
  M2 += delta * (x - mean);
}
const sampleVariance = count > 1 ? M2 / (count - 1) : NaN;
```

The three update statements are the entire recurrence. It processes observations in one pass, uses constant memory, and avoids recomputing deviations from a stored data array.

2 Exercise 2 – Sampling mean and variance

2.1 Chosen score distribution and theoretical moments

Choose a discrete uniform distribution over six possible scores:

$$X \in \{0, 2, 4, 6, 8, 10\}, \qquad P(X = x) = \frac{1}{6}.$$

Its theoretical mean is

$$\mu = \mathbb{E}[X] = \frac{0 + 2 + 4 + 6 + 8 + 10}{6} = 5.$$

The variance and fourth central moment are

$$\sigma^2 = \mathbb{E}[(X-5)^2] = \frac{25 + 9 + 1 + 1 + 9 + 25}{6} = \frac{35}{3} \approx 11.6667,$$
$$\mu_4 = \mathbb{E}[(X-5)^4] = \frac{625 + 81 + 1 + 1 + 81 + 625}{6} = \frac{707}{3} \approx 235.6667.$$

2.2 Experiment across hackers

For hacker j, let $X_{j1}, \ldots, X_{jn}$ be the n independent scores. Apply Welford's updates to this trajectory and obtain

$$\bar{X}_j = \frac{1}{n}\sum_{i=1}^{n} X_{ji}, \qquad S_j^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_{ji} - \bar{X}_j)^2.$$

Across m hackers, calculate

$$\overline{\bar{X}} = \frac{1}{m}\sum_{j=1}^{m} \bar{X}_j, \qquad \overline{S^2} = \frac{1}{m}\sum_{j=1}^{m} S_j^2.$$

Because the individual scores are sampled from the same distribution and the sample variance uses divisor $n - 1$,

$$\mathbb{E}[\overline{\bar{X}}] = \mu = 5, \qquad \mathbb{E}[\overline{S^2}] = \sigma^2 = \frac{35}{3}.$$

Thus both averages are unbiased estimators of their corresponding population parameters. A single finite simulation will generally not equal the theoretical values exactly: the differences are due to sampling (Monte Carlo) error. Increasing m makes the averages across hackers more stable.

2.3 Effect of increasing the number of systems n

For independent observations,

$$\mathrm{Var}(\bar{X}_j) = \frac{\sigma^2}{n}, \qquad \mathrm{SD}(\bar{X}_j) = \frac{\sigma}{\sqrt{n}} = \sqrt{\frac{35}{3n}}.$$

Consequently, the distribution of individual hackers' sample means becomes narrower as n increases. The standard error of the average across m hackers is smaller still:

$$\mathrm{SD}(\overline{\bar{X}}) = \frac{\sigma}{\sqrt{mn}}.$$

The sample variance is also a random statistic. For an i.i.d. sample with finite fourth central moment,

$$\mathrm{Var}(S_j^2) = \frac{1}{n}\left(\mu_4 - \frac{n-3}{n-1}\sigma^4\right), \qquad n \geq 2.$$

For the selected distribution, this becomes

$$\boxed{\mathrm{SD}(S_j^2) = \sqrt{\frac{1}{n}\left[\frac{707}{3} - \frac{n-3}{n-1}\left(\frac{35}{3}\right)^2\right]}}.$$

This standard deviation also decreases as n grows (at an asymptotic rate proportional to $1/\sqrt{n}$). Averaging the sample variances over m independent hackers reduces the standard deviation by a further factor of $\sqrt{m}$:

$$\mathrm{SD}(\overline{S^2}) = \frac{\mathrm{SD}(S_j^2)}{\sqrt{m}}.$$

2.4 Conclusions and relevance to inferential statistics

1. The average of the sample means should be close to the theoretical mean $\mu = 5$, and the average of the unbiased sample variances should be close to the theoretical variance $\sigma^2 = 35/3$. The match improves as the number of independent hackers m increases.

2. Increasing n, the number of systems attempted by each hacker, reduces the variability of both statistics. The sample mean concentrates at a rate $1/\sqrt{n}$; the sample variance also concentrates as the sample size grows.

3. These results are central to inferential statistics. In practice, population parameters are unknown, so they are estimated from samples. The sampling distribution describes how an estimator varies from sample to sample. Its spread provides standard errors and is the foundation for confidence intervals and hypothesis tests.

The HTML companion runs this Monte Carlo experiment for several user-selected values of n. Its table reports the across-hacker averages and the empirical standard deviations, alongside the theoretical targets and standard deviations. Small empirical differences from the theoretical quantities are expected for finite m.